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AlgosQuest 2025

Round 1 Submission by TEAM-zero



TEAM-Zero

Problem Statement : Compressed & Encrypted Data Transfer Protocol

Domain: Secure Communication, Systems Engineering & Edge AI.

**SmartStream – The Intelligent,
Bandwidth-Aware Secure Tunnel.**

Members :

- 1)Lalith kumar Vedium
- 2)Abhiram Alikatti
- 3)Tejavardhan Ande
- 4)Varshith Chada
- 5)Vishwanth Munikuntla



Problem Validation & Relevance

Problem :

In low-bandwidth environments (2G/Satellite/Rural India), file transfer is broken. Standard tools (WhatsApp/Email) fail because they rely on heavy central servers and inefficient "one-size-fits-all" compression.

The Gap:

- VPNs provide security but slow down speed.
- Standard Compression (Zip) is static; it doesn't adapt to the file's actual content.
- Cloud Transfer wastes bandwidth by uploading to a server before the receiver can download.

Proposed Solution Overview

The Solution:

- **An AI-Native P2P Protocol.** SmartStream moves the "Intelligence" to the client-side (Edge). It analyzes files before sending, applies the mathematical optimal compression, and streams directly to the receiver.

Core Innovation: "Content-Aware Adaptive Pipeline"

- Most tools just Zip. SmartStream uses a Nano-AI to scan file entropy and switch algorithms (Zstd vs Brotli vs LZ4) in real-time.

Key Differentiator:

- **Serverless Architecture.** We use WebRTC for a direct peer-to-peer pipe, meaning 0% server bandwidth cost and 100% data privacy (the file never touches our cloud).

Technical Architecture Overview

1. **Input Analysis:** The browser reads the file in 64KB Blocks. A lightweight Random Forest Classifier (running in TensorFlow.js) calculates entropy.

2. **Adaptive Compression:**

Text/Code → Brotli (High density).

Binary/Data → Zstandard (Zstd) (Balanced).

High-Entropy (Images) → No Compression (Avoids overhead)

3. **Security Layer:**

Key Exchange: ECDH (Elliptic Curve Diffie-Hellman) generates a shared secret without transmission.

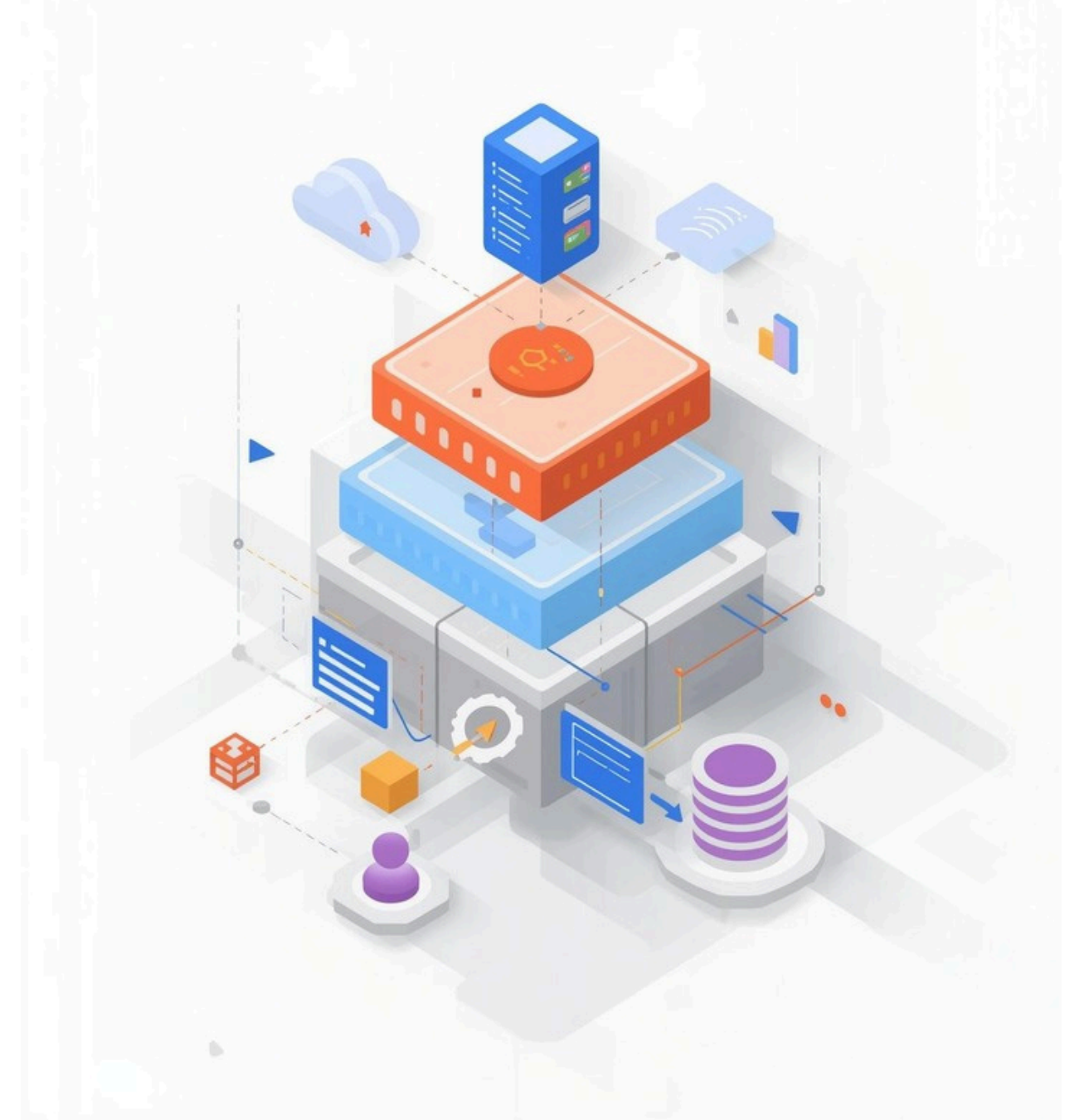
Encryption: ChaCha20-Poly1305. (Faster than AES on mobile/IoT devices).

4. **Transport Layer:**

Data is chunked and sent via **WebRTC Data Channels** (over SCTP for reliability), bypassing the server completely.

Tools and Technologies Overview

- Frontend (The Engine): React + Vite (UI), WebAssembly (WASM) (to run C++ compression code like Zstd in the browser at native speed).
- AI/ML: Scikit-Learn (Training), ONNX Runtime (Inference in Browser).
- Networking: PeerJS / Simple-Peer (WebRTC abstraction), Node.js + Socket.io (Signaling Server only).
- Security: Web Crypto API (Native browser standard).



Roadmap:

- MVP: P2P Transfer of Text/Images with Manual Key Sharing.
- Phase 2: Integration of Auto-ECDH Handshake and AI Selector.
- Phase 3: "Resumable Uploads" for broken connections.

Business Impact and Use Cases

- **ROI (Return on Investment):**
 - **Cost:** Near-zero infrastructure cost (since bandwidth is P2P).
 - **Efficiency:** AI compression saves an average of 30-40% bandwidth on mixed datasets compared to standard Zip.
- **Use Cases:**
 - **Defense/Military:** Securely sending maps/logs over poor radio links.
 - **Remote Healthcare:** Sending X-rays/Reports from rural clinics to city hospitals.
 - **Enterprise:** Internal secure document sharing (No cloud leaks).

Team Roles and Contributions

- 1) Lalith kumar Vedium
- 2) Abhiram Alikatti
- 3) Tejavardhan Ande
- 4) Varshith Chada
- 5) Vishwanth Munikuntla



Addressing Risks and Future Steps

- **Challenge:** Browser memory limits (RAM) when compressing huge files.
 - **Mitigation:** We implemented "Stream Processing"—reading, compressing, and sending small chunks one by one, so the whole file is never in RAM.
- **Challenge:** Firewall restrictions (NAT traversal).
 - **Mitigation:** Use public STUN/TURN servers to punch through firewalls.
- **Future Scope:**
 - Implement a custom UDP Congestion Control algorithm (like Google BBR) to further optimize speed on packet-loss networks.